21-S2-Q1

Q: (a) $px^2 + qy^2 = h$

objective function min.

(b) 12 bit

4 4 4

a b h

$p = \frac{a}{a+b}$

$q = \frac{b}{a+b}$

$S = ?$

$p \; q \; h \rightarrow \alpha_i ?$

Solution (a) Objective function

① For the perfect fit, we need the three given points to satisfy

$$p x_i^2 + q y_i^2 = h \quad , \quad i = 1,2,3$$

② we use the sum of squard residuals

$$f(\lambda) = \sum_{i=1}^{3} (p x_i^2 + q y_i^2 - h)^2 \quad \text{with } \lambda = (p, q, h)$$

③ An optimisation algorithm minimises $f(\lambda)$

$f=0$ means all three points lie exactly on the cupping profile.

(c) $P_1\,(12, 5.6)$

$P_2\,(7, 3.4)$

$P_3\,(-10, 4.5)$

calculate fitness

1000 0100 1011

0101 1001 1100

0100 1101 1110

(b) Genotype → Phenotype

① let denot $S = \langle \alpha_1 \alpha_2 \cdots \alpha_{12} \rangle$ with $\alpha_i \in \{0,1\}$

$$a = \sum_{j=1}^{4} \alpha_j 2^{4-j} \quad \in [0,15]$$

$$b = \sum_{j=5}^{8} \alpha_j 2^{8-j} \quad \in [0,15]$$

$$h = \sum_{j=9}^{12} \alpha_j 2^{12-j} \quad \in [0,15]$$

$$p = \frac{a}{a+b}$$

$$q = \frac{b}{a+b}$$

②

$$p = \frac{\sum_{j=1}^{4} \alpha_j 2^{4-j}}{\sum_{j=1}^{4} \alpha_j 2^{4-j} + \sum_{j=3}^{8} \alpha_j 2^{8-j}}$$

$$q = \frac{\sum_{j=3}^{8} \alpha_j 2^{8-j}}{\sum_{j=1}^{4} \alpha_j 2^{4-j} + \sum_{j=3}^{8} \alpha_j 2^{8-j}}$$

$$h = \sum_{j=9}^{12} \alpha_j 2^{12-j}$$

(c) ① P_1 (12, 5,6)

P_2 (7, 3.4)

P_3 (−10, 4.5)

②

ID	genotype	phenotype (a, b, h)	p q h	f(λ)	fitness
1	1000 0100 1011	8 4 11	$\frac{2}{3}$ $\frac{1}{3}$ 11	13658.4495	7.3209×10^{-5}
2	0101 1001 1100	5 9 12	$\frac{5}{14}$ $\frac{9}{14}$ 12	5067.2700	1.9731×10^{-4}
3	0100 1101 1110	4 13 14	$\frac{4}{17}$ $\frac{13}{17}$ 14	2590.3141	3.8590×10^{-4}

$$f(\lambda_1) = \sum_{i=1}^{3} (p x_i^2 + q y_i^2 - h)^2$$

$$= \left(\frac{2}{3}\times 12^2 + \frac{1}{3}\times 5.6^2 - 11\right)^2$$

$$+ \left(\frac{2}{3}\times 7^2 + \frac{1}{3}\times 3.4^2 - 11\right)^2$$

$$+ \left(\frac{2}{3}\times 10^2 + \frac{1}{3}\times 4.5^2 - 11\right)^2$$

$$= 13658.4495$$

$$f(\lambda_2) = 5067.2700$$

$$f(\lambda_3) = 2590.3141$$

③ we denote $\text{fitness} = \dfrac{1}{1+f(\lambda)}$

So the third string have the highest fitness.

(d) Binary-coded vs. real-valued

	Binary-coded	real-valued
precision	Limited to 16 discrete levels	Arbitary precision
Constraint	simple to embed $p+q=1$ via $p=a/(b+a)$ $q=b/(b+a)$	optimise only in $p+q=1$
Search space	$2^{12}=4096$ fixed candidates	continuous and infinite space
Genetic operators	standard bit crossover and mutation and representation independent	Arithmetic/blend crossover and Gaussian mutation
Implementation	easy to code	set step sizes handle boundaries.

② Ad. Disad.

(1) Binary coding is portable and operator independent, ideal for mixed or categorical variables.

(2) But resolution grows exponentially with needed precision

(3) Real-valued coding matches continuous design parameters directly, avoid discretisation error

(4) But it needing carefully designed real space variation operators and hyper parameters.